The Induction Principle — problem 8.36

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The positive integers $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$ are given. The sums $x_1 + \cdots + x_n$ and $y_1 + \cdots + y_m$ are equal and less than mn. Prove that one may cross out some of the terms in the equality $x_1 + \cdots + x_n = y_1 + \cdots + y_m$, so that one again gets an equality.